

**ARCH 594/508: AI Research Seminar/Studio—AI in Architecture**  
Spring 2026

# Data & Data in Architecture

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Sustainable Intelligence Lab [SIL]

# TODAY'S AGENDA

1. Data Definition, Types and Sources, Data in Architecture/ Built Environment
2. Lab Activity: Exploratory Data Analysis (EDA)

# DEFINITION OF DATA



Data refers to **information, facts, or observations** collected, stored, and processed for various purposes.



It can manifest in **various forms** such as words, numbers, measurements, observations, or symbols.



**Key Points:**

Raw material for deriving information and knowledge.

Foundation for decision-making, analysis, and research across diverse fields.

# DATA, INFORMATION, AND KNOWLEDGE

1. **Data:** Fundamental information unit, foundational for knowledge.
2. **Information:** Organized datasets with contextual connections.
3. **Knowledge:** Goes beyond data, emerges from patterns and connections.
4. **Wisdom:** Applies refined, contextual knowledge practically.

# TYPES OF DATA

- **Categorical (Qualitative) Data:**

- Observable but not measurable attributes: colors, textures, smells, tastes.

- **Numerical (Quantitative) Data:**

- Measurable attributes: speed, age, temperature, height.

## **Discrete Data:**

- Takes specific values (e.g., 1, 2, 3...).

## **Continuous Data:**

- Can take any value within a particular range (e.g., [3.1, 8]).

**Discrete data is counted;**  
**Continuous data is measured.**

# TEMPORAL AND SPATIAL DIMENSIONS OF DATA

## Temporal/ Time

### ■ **Static Data:**

- No variation over time.
- Example: Annual mean temperature in Seattle in 2023.

### ■ **Dynamic Data** (Time Series or Temporal):

- Varied data over time.
- Example: Hourly relative humidity in Seattle during January 2024.

# TEMPORAL AND SPATIAL DIMENSIONS OF DATA

## Spatial/Location

### ■ **Non-Geospatial Data:**

- May or may not include geo-referenced information.
- Cannot be mapped due to formatting or lack of geographic coordination information.

### ■ **Geospatial Data:**

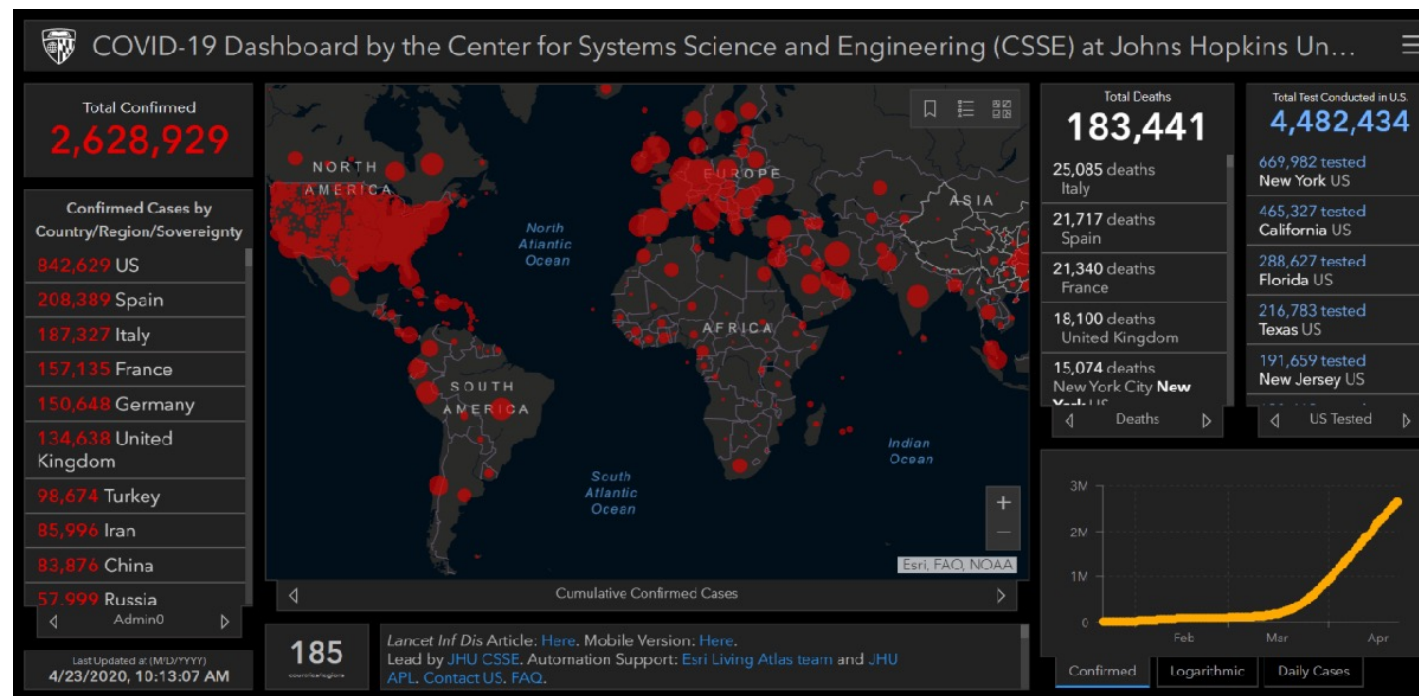
- Includes geo-referenced information, mappable data.
- **Types:** Point-Location, Zonal Boundaries, Hypothetical Boundaries, Spatiotemporal.

# GEOSPATIAL DATA TYPES

- Point-Location Geospatial Data
  - Zonal Boundaries Geospatial Data
  - Zonal Boundaries (Inner City) Geospatial Data
  - Hypothetical Boundaries Geospatial Data
- 
- **Spatiotemporal (Geospatial + Temporal)**

## EXAMPLE: GEOSPATIAL DATA

- Point Location



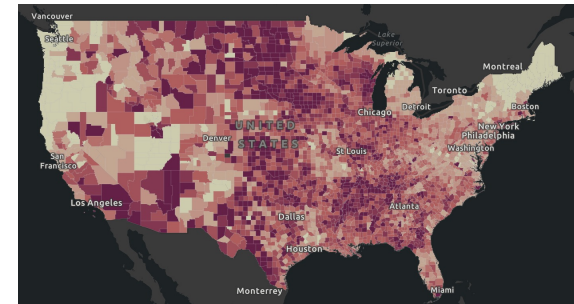
Example: Point Location; Courtesy of John Hopkins University

# EXAMPLE: GEOSPATIAL DATA

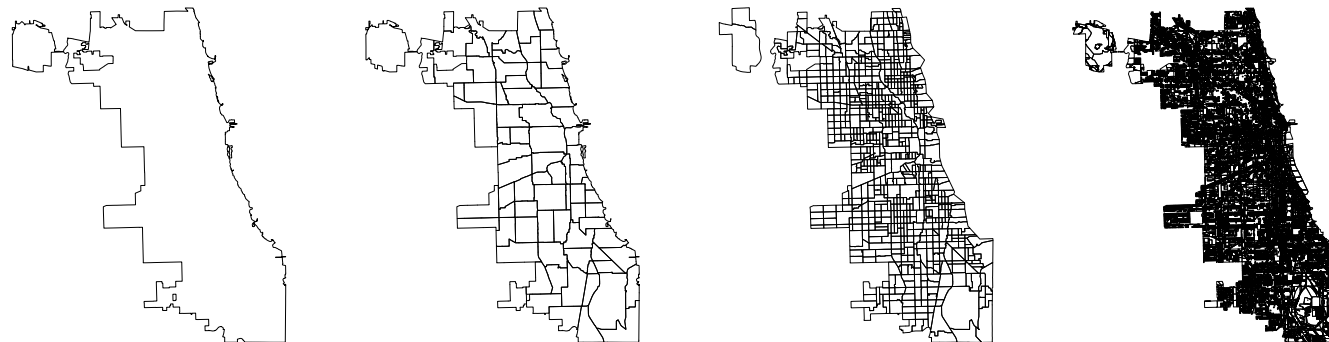
## ■ Zonal Boundaries



State



County



Example: Zonal Boundaries

# PYTHON DATA TYPES (CORE CONCEPTS)

- **Numeric:**

int (integers /  $x = 10$ ), float (decimals /  $y = 3.14$ ), complex (complex numbers /  $z = 2 + 3j$ )

- **Sequence (ordered):**

List (lst = [1, 2, 3]) , tuple (t = (1, 2, 3)) , range (r = range(5))

- **Text:**

str — strings for textual data ( name = "AI Studio")

- **Boolean:**

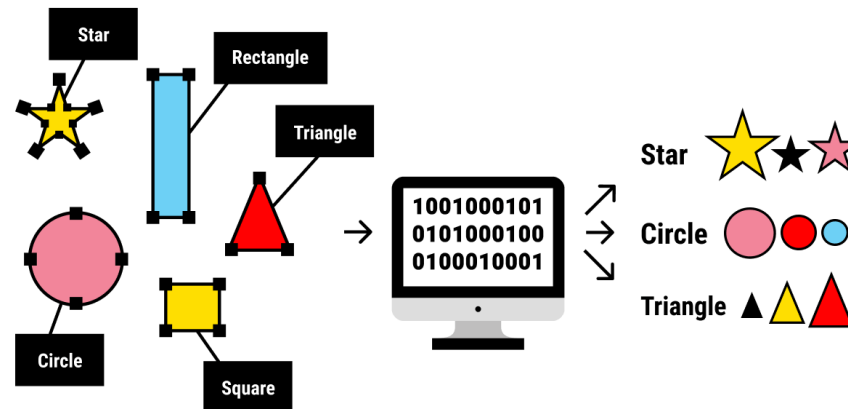
bool — True / False for logic and conditions (is\_valid = True)

# FORMATTING OF DATASETS

- Non-Geospatial Datasets:
  - Examples: CSV, XLSX, JSON, RDF, XML, etc.
- Geospatial Datasets:
  - Examples: Shapefiles, KML, KMZ, GeoJSON.

# LABELED AND UNLABELED DATA

- **Labeled Data:** This data has both input features and their known outcomes. For example, in an image dataset, each picture is labeled with what it contains, like "cat" or "dog."
- **Unlabeled Data:** This data only includes input features without known outcomes. For instance, it lacks labels like "cat" or "dog" for each image.



# DATA IN BUILT ENVIRONMENT



**Structured Databases:** Stored in tables for easy querying and analysis; public data



**Unstructured Data (Text):** Extracting insights from social media, emails, news



**Big Data:** Generated from various sources such as Public Wi-fi, Smartphone Apps, etc.)



**Sensor Readings:** IoT sensor data (e.g., temperature, occupancy, mobility) for monitoring environments, cities, industries, and healthcare.



**Simulated Data:** Physics-based engineering simulations.

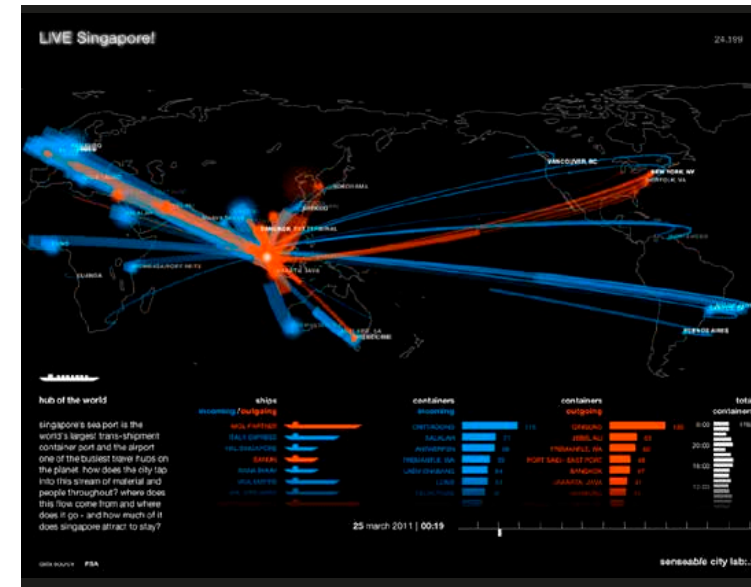
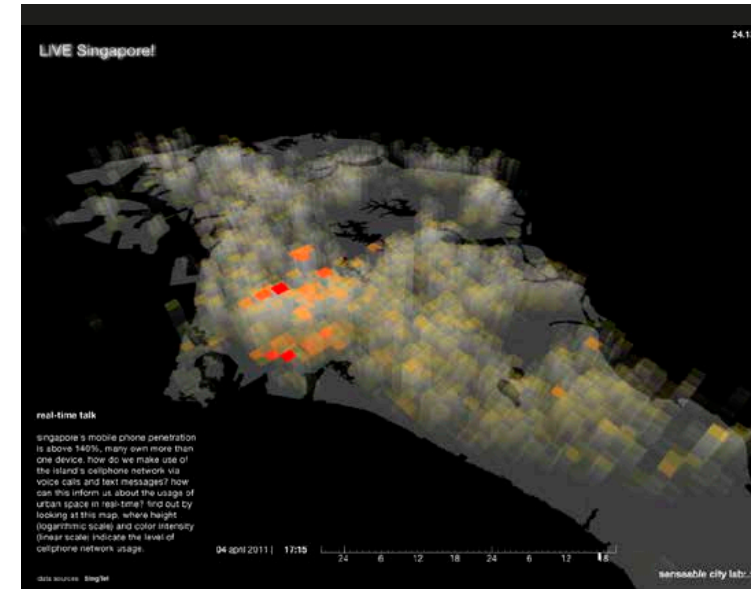
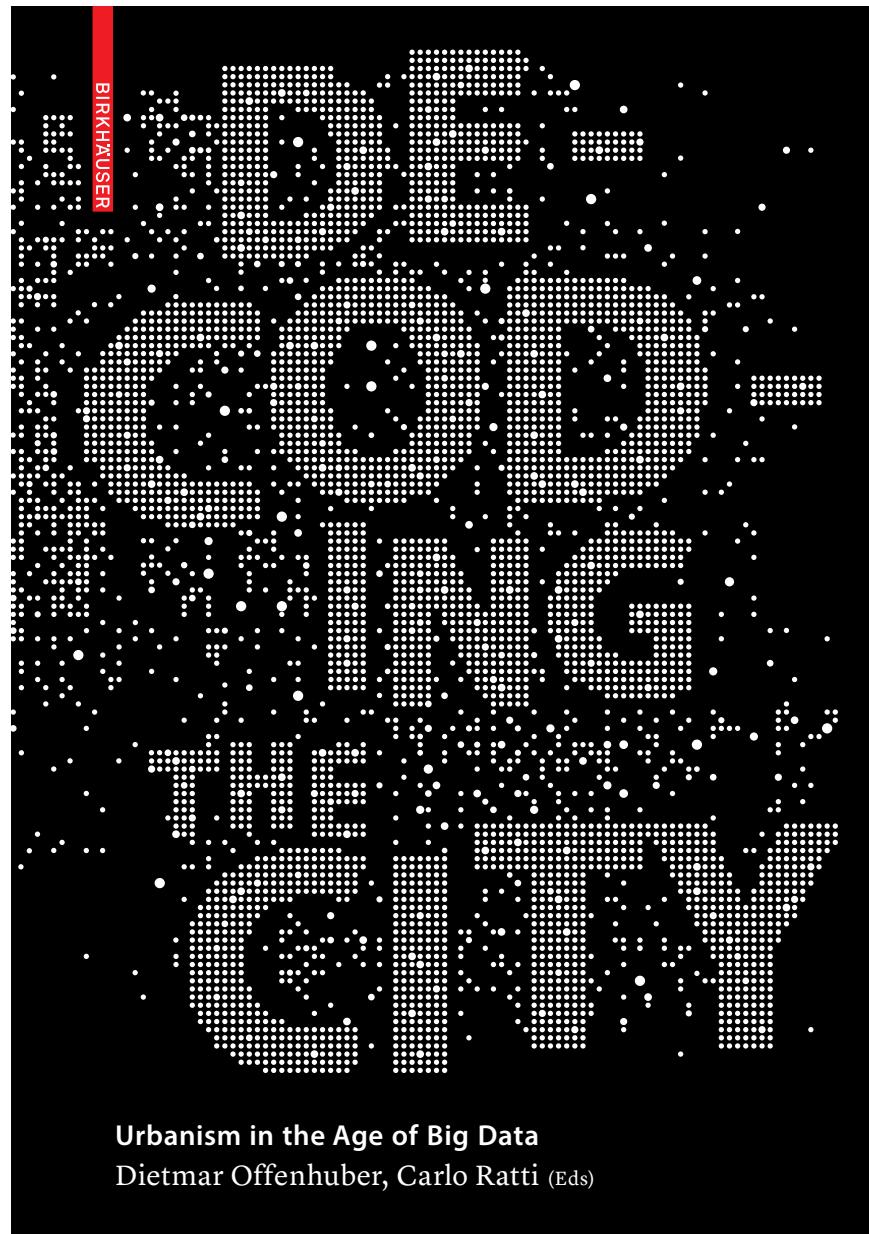


**Other Generated Data:** Derived from various design layouts and scenarios.

# BIG URBAN DATA



@pexels.com

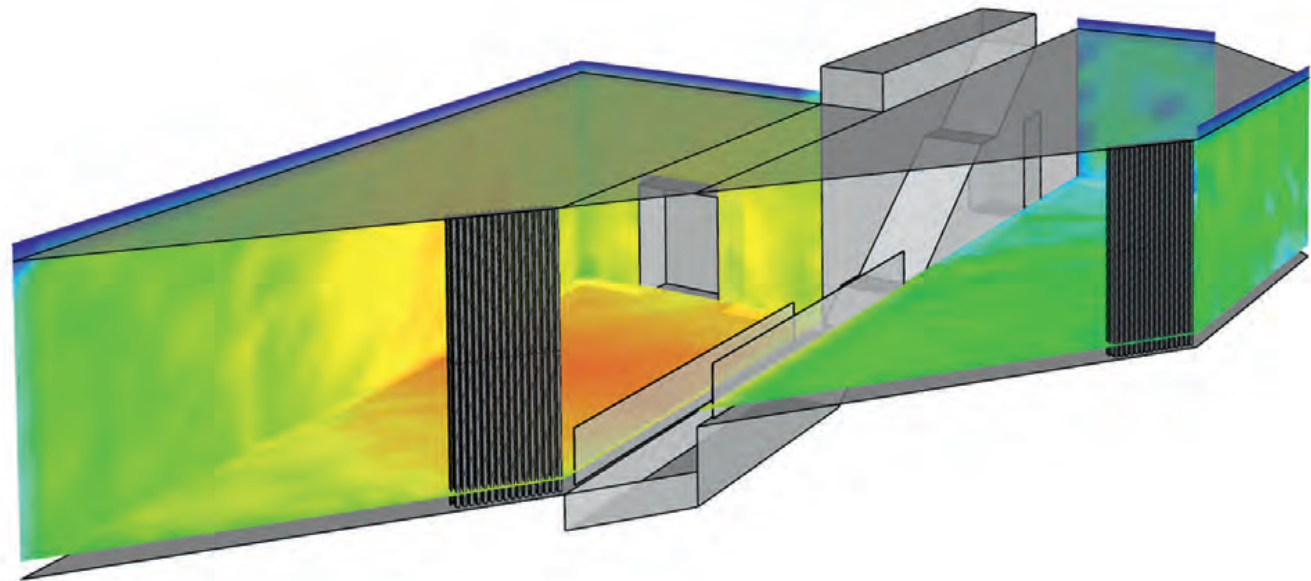


# DATA FROM SENSORS

- Daylight redirection system, connecting digital tools to real-time monitoring
- Sensors deployed in the office cubicle and the prototype physical layer.
  - They collect data on temperature, humidity, light, motion, CO2 and send it via Wi-Fi.

# DATA FROM SIMULATIONS

- simulation data / synthetic data

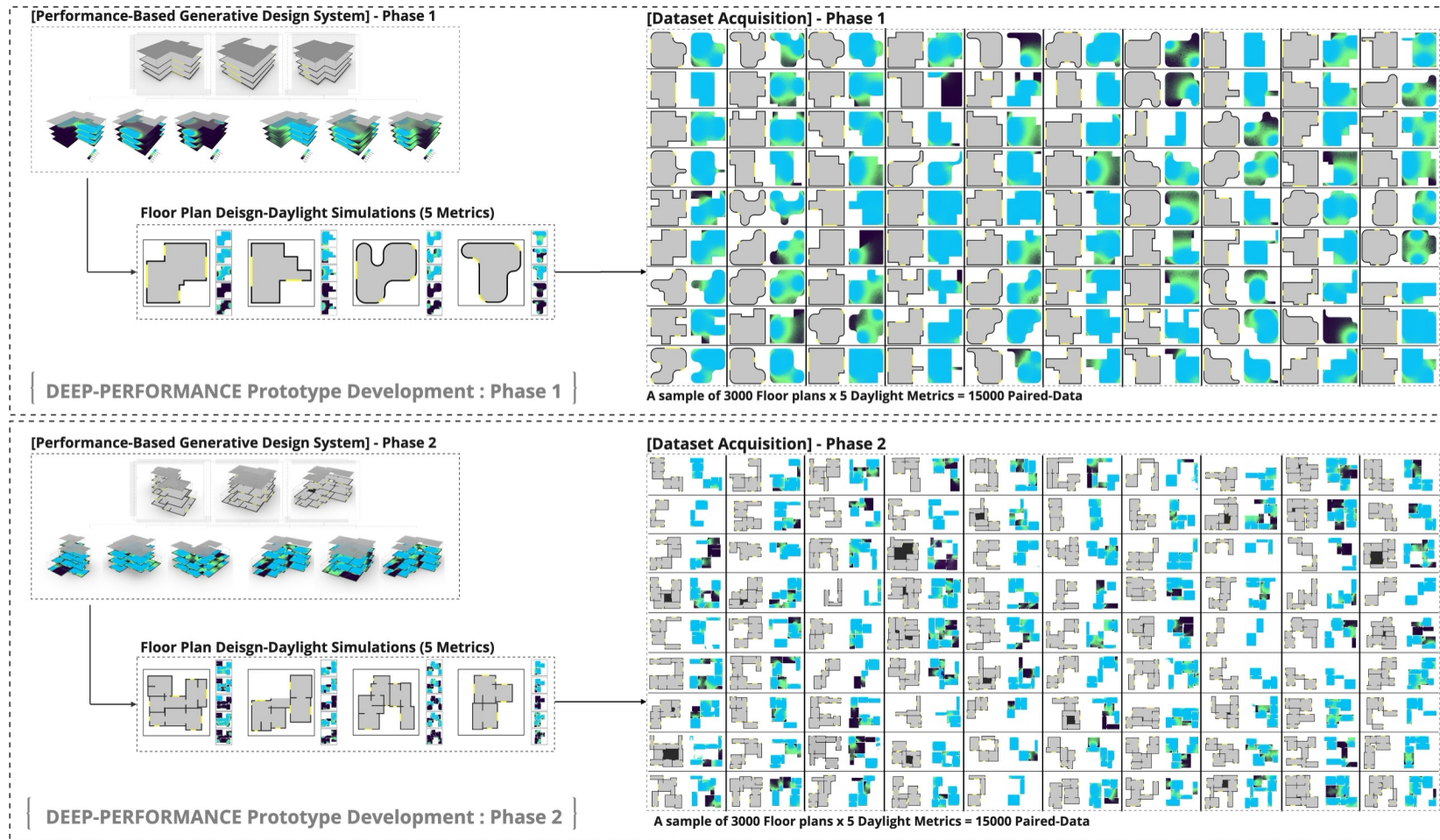


Peters, Brady, and Terri Peters. (2018).

## ■ simulation data

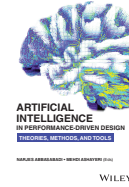


Yousif, Sh., Bolojan, D. (2024). An Integrative Deep Performance Framework for Early Design Ideation. *Artificial Intelligence in Performance-driven Design: Theories, Methods, and Tools*, edited by Abbasabadi, N., Ashayeri, M. Wiley.

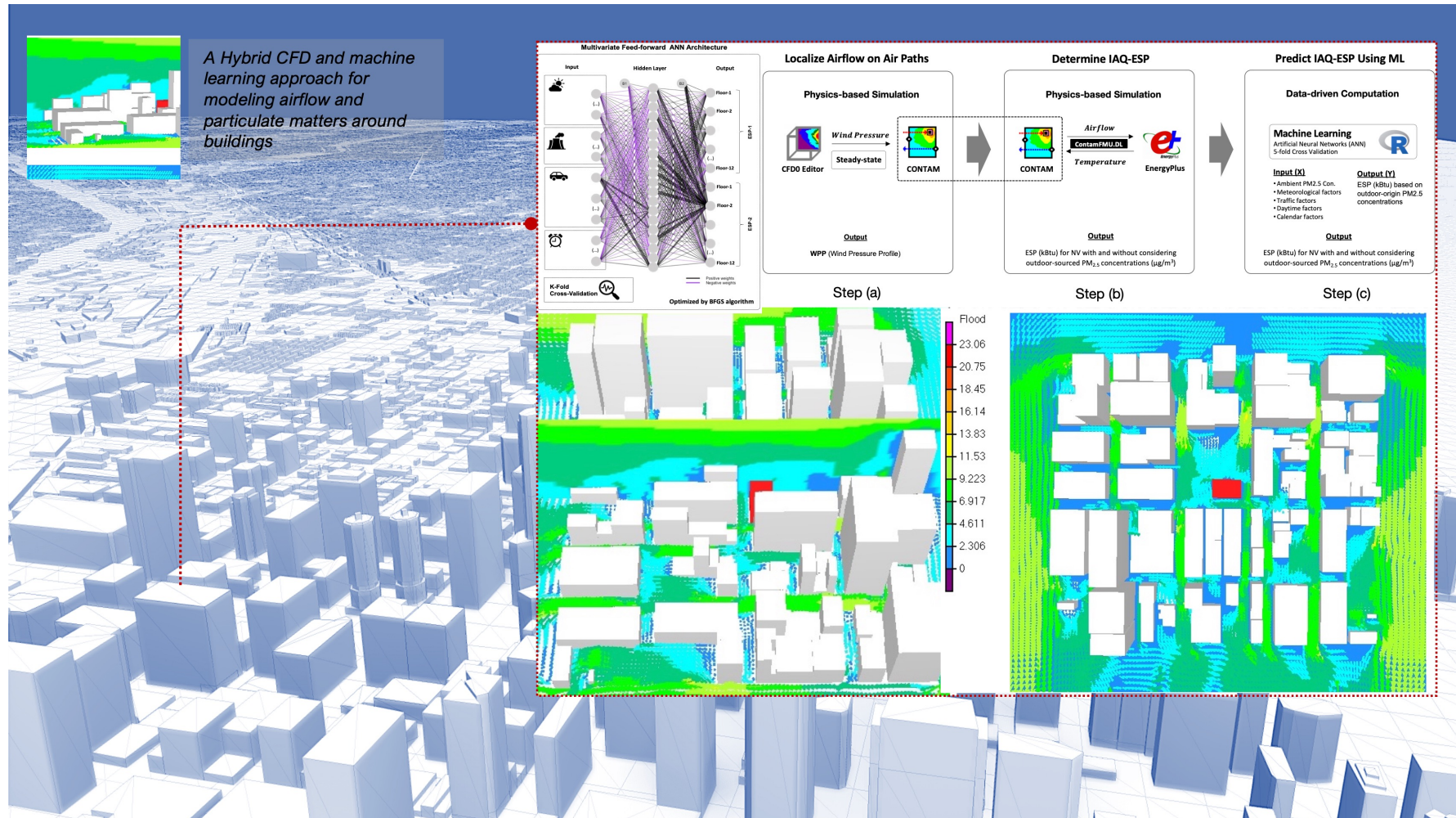


The datasets employed to train the surrogate model for both Phases 1 and 2 are generated from a parametric system, which integrates a daylight simulation of five key metrics.

## ■ simulation data



Ashayeri, M., Abbasabadi, N. (Forthcoming). Hybrid Physics-based machine learning approach for multi domain modeling. *Artificial Intelligence in Performance-driven Design: Theories, Methods, and Tools*, edited by Abbasabadi, N., Ashayeri, M. Wiley.

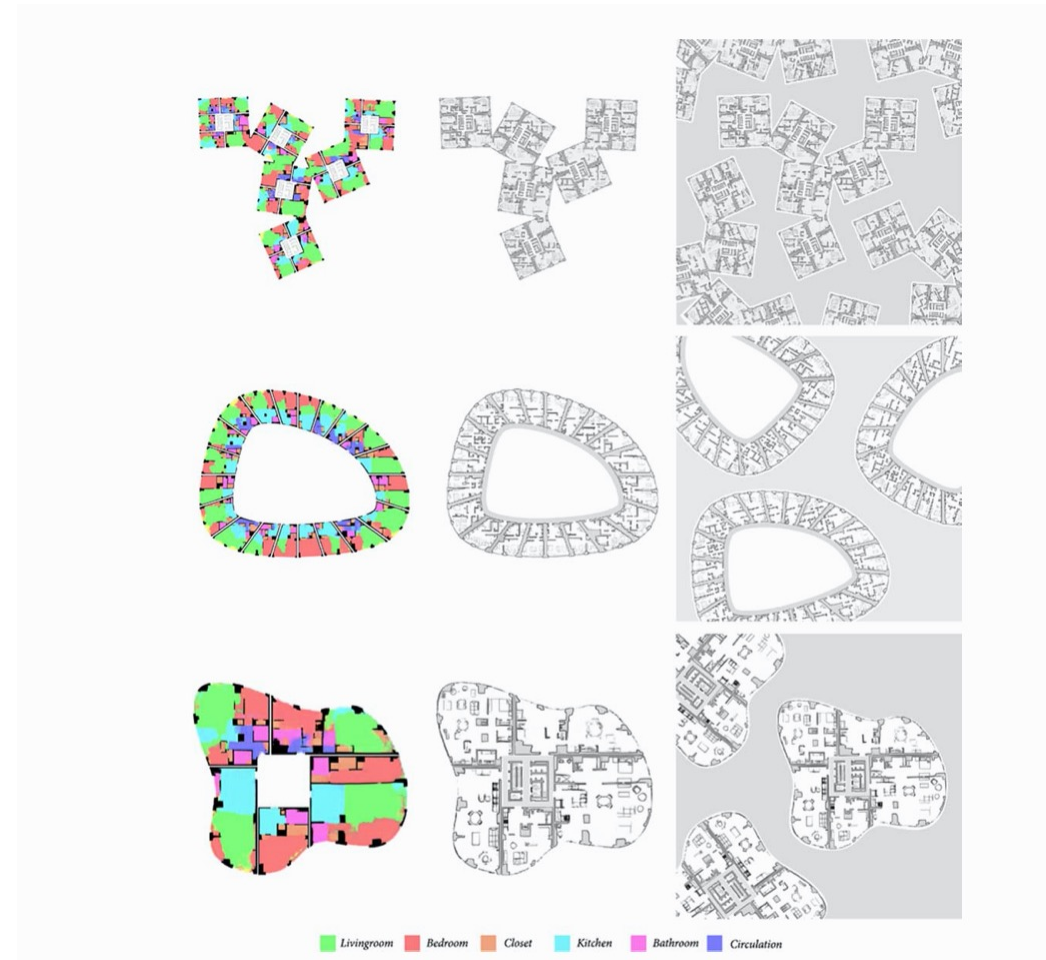


# OTHER GENERATED DATA

**Derived from various design layouts and scenarios**

## ArchiGAN

- A version of GANs that uses a Pix2Pix GAN-model to design floor plans for an entire building
- A database of over 800 annotated plans is used as input, and as output the system generates a layout with rooms encoded with colors to specify a program.



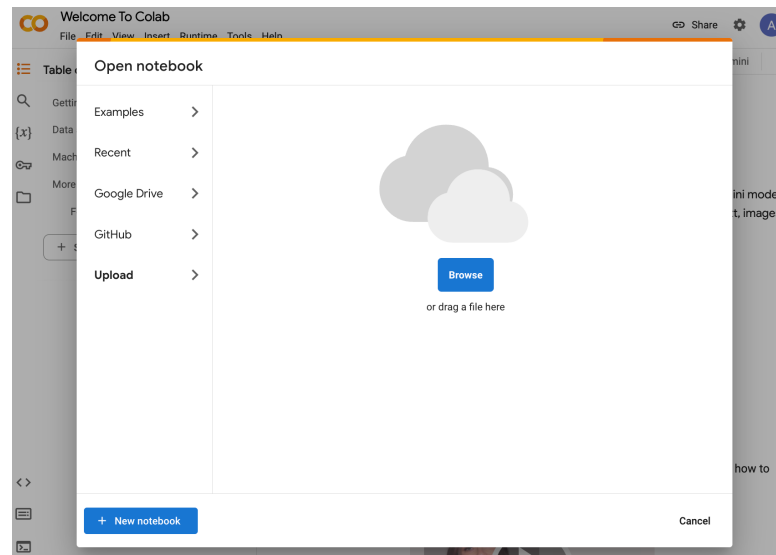
Stanislas Chaillou, ArchiGANs. Chaillou is able to generate stacked plans of buildings using ArchiGAN, a Pix2Pix version of GANs.

# SUMMARY

- Data: Data definition, Data types & Data in Architecture

# LAB ACTIVITY: EXPLORATORY DATA ANALYSIS

1. Open up a web browser. Google “google colab” click on the first entry. You may also access the website by following <https://colab.research.google.com/>
2. Go to Canvas and download the file called “Exploratory\_Data\_Analysis\_Pandas.ipynb”



# NEXT CLASS

- Exploratory Data Analysis

